

E-Commerce Sales & Customer Analytics

Data Cleaning, SQL Analysis & Business Insights



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Data Analyst

Tools Used:

Excel | Power Query | MySQL | SQL | Power BI

Dataset: E-Commerce Sales Transactions

Project Overview

The **E-Commerce Sales & Customer Analytics** project analyzes transactional e-commerce data to understand **sales performance, customer behavior, product performance, order trends, and operational performance**.

The project follows an end-to-end data analytics workflow. The raw dataset was cleaned and transformed using **Excel and Power Query**, including handling missing values, duplicates, inconsistent formats, and data-standardization issues. The cleaned dataset was then imported into **MySQL Workbench** for business-focused SQL analysis.

The analysis covers **revenue, orders, customers, products, categories, discounts, payment methods, regions, delivery performance, and customer behavior**. SQL techniques including **CTEs, subqueries, CASE statements, aggregations, and window functions** are used to derive meaningful business insights.

The results are presented through **Excel visualizations and an interactive Power BI dashboard**, highlighting important KPIs, trends, comparisons, and actionable findings.

Project Workflow

Raw Data → Power Query → Clean Dataset → MySQL → SQL Analysis → Excel Charts → Power BI Dashboard → Insights & Recommendations

Tools Used

Excel | Power Query | MySQL Workbench | SQL | Power BI

Key Analytical Areas

- Revenue and sales performance
- Order and customer analysis
- Product and category performance
- Discount and pricing analysis
- Payment and regional performance
- Delivery and customer behavior
- Business KPIs and trends

Objective

To transform transactional data into **clear, reliable, and actionable business insights** that support better decisions related to **revenue growth, customer engagement, product performance, and overall business performance**.

Business Problem

E-commerce businesses generate large volumes of transactional data across customers, products, orders, pricing, payments, and delivery operations. However, raw transaction data does not directly provide a clear view of overall business performance or the factors influencing sales and customer activity.

The business needs to understand **how sales are performing, where revenue is being generated, which products and categories are driving demand, and how customers interact with the business**. It is also important to understand order outcomes, discount patterns, regional performance, payment preferences, and delivery performance to identify areas of strength and opportunities for improvement.

The analysis therefore addresses the following business questions:

- How is revenue performing over time, and which periods generate the highest Sale
- Which cities, regions, categories, and products contribute most to revenue?
- What is the overall order performance across delivered, cancelled, returned, and pending orders?
- How many new and repeat customers does the business acquire, and which customers generate the most revenue?
- How does customer purchasing behavior vary across the business?
- Which products and categories have the strongest sales performance?
- How do discounts relate to revenue and average order value?
- Which payment methods contribute most to orders and revenue?
- How does delivery performance vary across cities and categories?
- Which business areas present opportunities for improving revenue, customer engagement, and operational performance?

The project uses transactional data to answer these questions through structured SQL analysis and data visualization, providing a clearer view of the business and supporting data-driven decision-making.

Analysis Sections

The SQL analysis is organized into focused business areas to provide a structured view of e-commerce performance. Each section uses SQL-based analysis to answer specific business questions and is supported by relevant visualizations where appropriate.

1. Executive KPI Analysis

Evaluation of key business metrics including **revenue, orders, customers, quantity sold, AOV, ASP, and discount performance**.

2. Revenue Analysis

Analysis of **monthly revenue trends, revenue growth, city performance, category contribution, and top revenue-generating customers**.

3. Order & Business Performance

Evaluation of **delivered, cancelled, returned, and pending orders**, including order performance and monthly order trends.

4. Customer Analysis

Analysis of **first purchases, new customer acquisition, repeat customers, customer revenue, and customer-level purchasing behavior**.

5. Product & Category Analysis

Evaluation of **top-selling products, product revenue, category performance, quantity sold, ASP, and discount levels**, including category-based product rankings.

6. Discount & Pricing Analysis

Analysis of **discounted versus non-discounted orders, discount bands, revenue performance, and products with relatively high discounts**.

7. Delivery & Operational Analysis

Evaluation of **average delivery time, delivery performance by city and category, and operational trends**.

8. Advanced Customer & SQL Analysis

Application of advanced SQL techniques to analyze **customer cumulative revenue, regional customer rankings, and purchase intervals** using window functions such as LAG(), RANK(), and DENSE_RANK()

Analytical Approach

The analysis combines **business metrics, aggregation, date-based analysis, customer-level analysis, rankings, comparisons, CTEs, subqueries, and window functions** to convert transactional data into meaningful business findings.

Dataset Description

The project uses an **e-commerce transactional dataset** containing order-level information related to customers, products, sales, payments, discounts, order status, and delivery. The dataset provides a consolidated view of customer transactions and supports analysis of sales performance, customer purchasing behavior, product performance, and operational trends.

Each row represents a **transactional order record**, with information covering the customer, location, order and delivery dates, product details, quantity, pricing, discount, payment method, order status, and customer rating.

Dataset Fields

Field	Description
Order_ID	Unique identifier of the order
Customer_ID	Identifier of the customer
Customer_Name	Name of the customer
Age	Customer age
City	Customer city
State	Customer state
Region	Geographic region
Order_Date	Date the order was placed
Delivery_Date	Date the order was delivered
Product_ID	Identifier of the product
Product_Name	Name of the product
Category	Product category
Brand	Product brand
Quantity	Number of units ordered
Selling_Price	Selling price of the product
Discount_%	Discount percentage applied
Payment	Payment method used
Order_Status	Status of the order
Rating	Customer rating for the order

Dataset Scope

The dataset supports analysis across several dimensions, including **time, geography, customers, products, categories, pricing, payments, and order status**. These fields allow transactional data to be examined from both customer and business perspectives.

The dataset is used as the primary source for the project's **SQL analysis and Power BI dashboard**, while selected analysis results are also visualized using Excel charts.

Analytical Use

The dataset enables the project to investigate **revenue trends, order performance, customer behavior, product and category performance, discount patterns, regional performance, payment preferences, and delivery-related metrics**.

This structure provides the foundation for transforming transactional records into **business-focused analysis and actionable insights**.

Dataset Summary

Attribute	Details
Total Rows	10,000
Total Columns	19
Data Type	E-commerce transactional data
Primary Unit	Order/transaction record
Source	E-commerce sales dataset

Data Cleaning & Quality

he dataset was systematically reviewed and cleaned using **Excel and Power Query**. Each column was assessed for missing values, blanks, duplicates, inconsistent formatting, invalid values, and data-type issues. Cleaning decisions were based on the business meaning of each field.

Duplicate Records

- Found duplicate Order_ID records
- Identified **77 duplicate Order_ID records**.
- Removed duplicate records.
- Final dataset reduced to **9,923 records**.

Customer_ID

- Identified **9 missing values**.
- Replaced missing IDs with Unknown because the customer identifier could not be reliably reconstructed.

Customer_Name

- Identified missing/null and blank values.
- Applied **Trim/Clean** for text consistency.
- Missing names were handled as Unknown.
- Avoided blindly applying Proper Case because customer names may require controlled formatting.

Age

- Identified **110 missing values**.
- No significant outliers were identified during validation.
- Missing values were handled using the **median age of 42**

City

- Identified missing and blank values.
- Applied **Trim/Clean** for text consistency.
- Missing values were replaced with Unknown.
- Standardized inconsistent city names, including:
 - Bangalore / Bengaluru
 - Calcutta / Kolkata
 - Hyderabad / Hyderabad
 - New Delhi / Delhi
- Applied consistent text formatting after standardization.

State

- No major structural issues remained after validation.
- Applied **Trim/Clean** for consistency.
- Missing values were handled as Unknown.
- Standardized text representation across state values.

Region

- No missing or blank values were identified.
- Applied **Trim/Clean** for consistency.
- Applied **Proper Case** to standardize text representation.
- Column was validated for consistent regional values.

Order_Date

- Identified multiple date formats within the column.
- Standardized the different date representations.
- Converted the column to the **Date** data type.
- Validated the transformed column.
- Final validation showed **84 valid values, 0 errors, and 16 empty values** during the intermediate validation stage, with the date values confirmed as correct after review.

Delivery_Date

- Identified missing delivery dates.
- Missing values were associated mainly with cancelled and returned orders, but **283 Delivered orders also had missing Delivery_Date values.**
- Did not create artificial delivery dates for missing records.
- Standardized the date values and converted the column to the **Date** data type.
- Final validation showed **84% valid values, 0% errors, and 16% empty values.**
- The missing Delivered-order dates were retained as a data limitation rather than being artificially filled.

Product_ID

- Reviewed the column for missing and blank values.
- Validated product identifiers for consistency.
- Applied text cleaning where required.
- Preserved Product_ID as the reference key for product-level analysis and lookup operations.

Product Name

- Identified **73+ missing or blank product names**.
- A significant portion of the missing names belonged to Delivered orders.
- Created a product lookup using Product_ID.
- Merged the lookup with the main dataset to recover missing product names.
- Matched records using Product_ID as the lookup key.
- Validated the resulting Product_Name column.
- The cleaned column contained **no remaining errors or missing values** after the lookup process.

Category

- No missing or blank values were identified.
- Applied **Trim/Clean** for consistency.
- Standardized inconsistent category naming.
- Changed Home and Kitchen to **Home & Kitchen**.
- Validated the standardized category values.

Brand

- No major missing-value issue was identified.
- Applied **Trim/Clean** for consistency.
- Reviewed brand values for inconsistent text representation.
- Preserved the intended brand naming rather than unnecessarily changing capitalization.

Quantity

- No missing values were identified.
- Identified negative quantities such as -1, -2, and -3.
- Most negative values were associated with Delivered orders, so the values were treated as sign inconsistencies rather than automatically removing the records.
- Applied **Absolute Value (ABS)** to correct negative quantities.
- Identified **22 records with Quantity = 0**.
- Zero quantities were retained because they were observed values and were not arbitrarily replaced

Selling Price.

- No missing values were identified.
- No zero or negative selling prices were identified.
- Values contained excessive decimal precision.
- Rounded values to an appropriate decimal precision for financial reporting.
- Standardized the column for consistent numerical analysis.

Discount%

- No missing-value issue was identified.
- Identified valid 0% discount values.
- Retained 0% because a product/order can legitimately have no discount.
- Identified negative discount values such as -0.05.
- Negative values were treated as sign inconsistencies.
- Applied **Absolute Value (ABS)** to correct negative discount values.
- Preserved valid decimal discount values after correction.

Payment Method

- Identified inconsistent payment-method naming.
- Applied **Trim/Clean** for text consistency.
- Standardized payment values:
 - Cash on Delivery → COD
 - Netbanking → Net Banking
 - Upi → UPI
- Identified null and blank values.
- Replaced missing and blank payment methods with Unknown.
- Validated the final payment categories for consistency.

Order Status

- Reviewed order-status values for consistency.
- Validated the available status categories used for business analysis.
- Preserved meaningful statuses such as **Delivered, Cancelled, Returned, and Pending**.
- Ensured consistent status naming for SQL-based order analysis.

Rating

- Reviewed the column for missing and blank values.
- Validated the rating values for consistency.
- Preserved available customer ratings without artificially assigning ratings to records where no rating was provided.
- Prepared the column for customer-rating analysis.

Data Validation

After completing the data cleaning and transformation process, the dataset was validated to ensure that the cleaned data was **consistent, structurally correct, and suitable for SQL analysis**.

Validation was performed in Power Query and Excel by reviewing data types, missing values, standardized values, numerical fields, date fields, and key business relationships.

Validation Checks

Record & Structure Validation

- Verified the final number of records after duplicate handling.
- Confirmed that the dataset contained the expected columns and structure.
- Checked that no unintended columns or transformation errors were introduced during the cleaning process.
- Confirmed that the cleaned dataset could be exported successfully for SQL analysis.

Data Type Validation

- Confirmed that date columns such as Order_Date and Delivery_Date use the **Date** data type.
- Confirmed that Quantity is stored as a numeric field.
- Confirmed that Selling_Price and Discount_% are stored as appropriate numeric data types.
- Confirmed that categorical fields such as City, State, Region, Category, Brand, Payment, and Order_Status contain text values.

Value & Consistency Validation

- Verified standardized city, state, region, category, brand, and payment values.
- Confirmed that negative Quantity and negative Discount_% values had been corrected.
- Confirmed that valid zero values, such as 0% discount and zero quantity records, were not removed without business justification.
- Checked that Selling_Price values were consistently rounded for financial analysis.
- Reviewed Order_Status values to ensure consistent categories.

Date Validation

- Confirmed that Order_Date values were converted to a consistent date format.
- Checked the relationship between Order_Date and Delivery_Date.
- Reviewed missing delivery dates separately rather than creating assumed dates.
- Confirmed that the date fields were suitable for monthly, yearly, and delivery-time analysis.

Final Validation Outcome

The validation process confirmed that the cleaned dataset was **consistent, structured, and ready for downstream analysis**. The validated dataset was then saved as a **CSV file for MySQL Workbench** and retained as an **Excel file for documentation and visualization**.

The validated dataset serves as the single prepared source for the project's **35 SQL analyses, Excel visualizations, and Power BI dashboard**.

SQL Analysis

After completing data cleaning and validation, the cleaned dataset was imported into **MySQL Workbench** for business-focused analysis. A set of **35 SQL questions** was developed to evaluate key aspects of e-commerce performance and customer behavior.

The analysis uses SQL techniques including **aggregations, CASE WHEN, date functions, CTEs, subqueries, and window functions such as LAG(), RANK(), and DENSE_RANK()** to answer business questions and identify meaningful patterns.

The analysis is organized into the following sections:

1. **Executive KPI Analysis**
2. **Revenue Analysis**
3. **Order & Business Performance**
4. **Customer Analysis**
5. **Product & Category Analysis**
6. **Discount & Pricing Analysis**
7. **Delivery & Operational Analysis**
8. **Advanced Customer & SQL Analysis**

Each analysis follows a consistent approach:

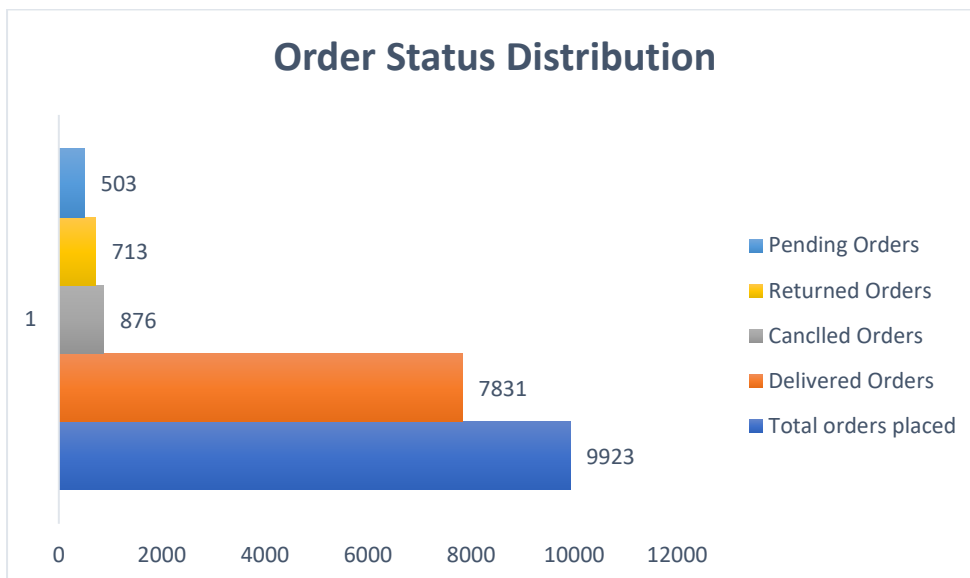
Business Question → SQL Query → Result → Visualization → Key Insight → Business Recommendation

The SQL results form the analytical foundation for the **Excel visualizations and Power BI dashboard** presented later in the project.

Executive KPI Analysis

1. Order Performance KPIs

KPI	Result
Total Orders Placed	9,923
Delivered Orders	7,831
Cancelled Orders	876
Returned Orders	713
Pending Orders	503



Insight

Out of **9,923 orders**, **7,831 were successfully delivered**, meaning the majority of orders reached customers. However, **876 orders were cancelled and 713 were returned**, resulting in **1,589 unsuccessful order outcomes**. In addition, **503 orders remain pending**.

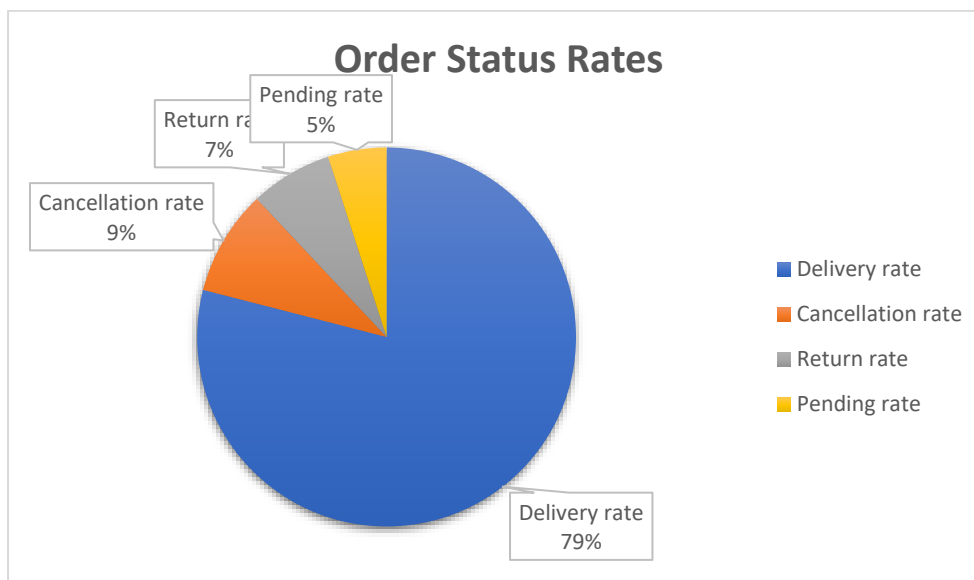
This means the business has a meaningful opportunity to improve order fulfillment beyond simply increasing order volume.

Recommendation

Prioritize an **order-loss analysis** for the **1,589 cancelled and returned orders**. Break them down by product, category, region, payment method and customer segment to identify the largest contributors. At the same time, monitor the **503 pending orders** to prevent fulfillment delays from converting into cancellations.

2. Order Status Rate KPIs

KPI	Result
Delivery Rate	78.92%
Cancellation Rate	8.83%
Return Rate	7.19%
Pending Rate	5.07%



Insight

The **78.92% delivery rate** shows that nearly four out of every five orders are successfully delivered. However, the **8.83% cancellation rate and 7.19% return rate together represent 16.02% of all orders**, which is considerably higher than the **5.07% pending rate**.

Therefore, cancellations and returns are the more significant operational concern compared with the current pending backlog.

Recommendation

Focus improvement efforts first on **cancellation and return reduction** rather than treating all unsuccessful statuses equally. Analyze the reasons and patterns behind the **16.02% combined cancellation and return rate**, then target the highest-impact products, locations and operational stages.

3. Core Business KPIs

KPI	Result
Total Quantity Sold	17,214
Unique Customers	3,988
Average Order Value	₹20,710.68

Insight

The business generated **17,214 delivered units from 3,988 unique customers**, while the average delivered order generated **₹20,710.68**. This indicates a relatively high order value compared with the unit-level selling price identified in the pricing KPIs.

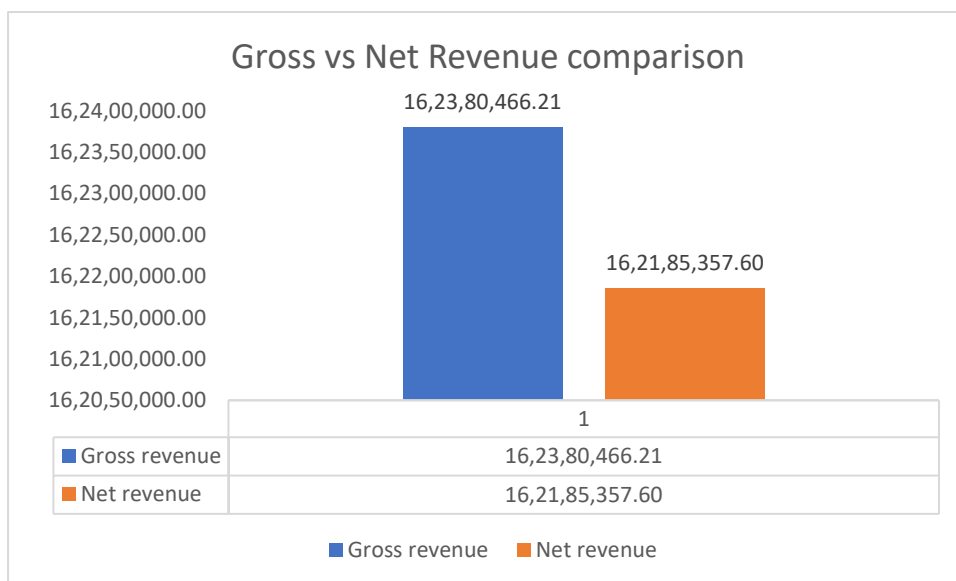
The combination of **17,214 units and 3,988 customers** also indicates that the business is generating multiple units across its customer base rather than relying exclusively on single-unit purchases.

Recommendation

Protect this existing basket-size behavior by introducing **cross-selling, product bundles and complementary-product recommendations**. These strategies can increase order value while avoiding unnecessary blanket discounts.

4.Revenue Performance KPIs

KPI	Result
Gross Revenue	₹162,380,466.21
Discount Amount	₹195,108.61
Net Revenue	₹162,185,357.60



💡 Insight

Delivered orders generated **₹162.38 million in gross revenue** and **₹162.19 million in net revenue**. Only **₹195,108.61** was given up through discounts, meaning the discount amount represents approximately **0.12% of gross revenue**.

The very small gap between gross and net revenue indicates that **discounting has a minimal impact on realized revenue in this dataset**.

🎯 Recommendation

Avoid increasing discounts broadly simply to stimulate sales. Instead, use **targeted promotional campaigns** for specific products, customer segments or periods where there is evidence that discounts can increase conversion or order frequency without unnecessarily reducing revenue.

5. Pricing Performance KPIs

KPI	Result
Average Selling Price (ASP)	₹9,421.71
Average Discount	0.12%

Insight

The average net selling price per unit is **₹9,421.71**, while the average discount is only **0.12%**. This indicates that the business is currently maintaining its selling price with **very limited reliance on discounting**.

When considered alongside the **₹20,710.68 AOV**, the average order generates substantially more revenue than the average unit price, suggesting that customers are purchasing multiple units per delivered order.

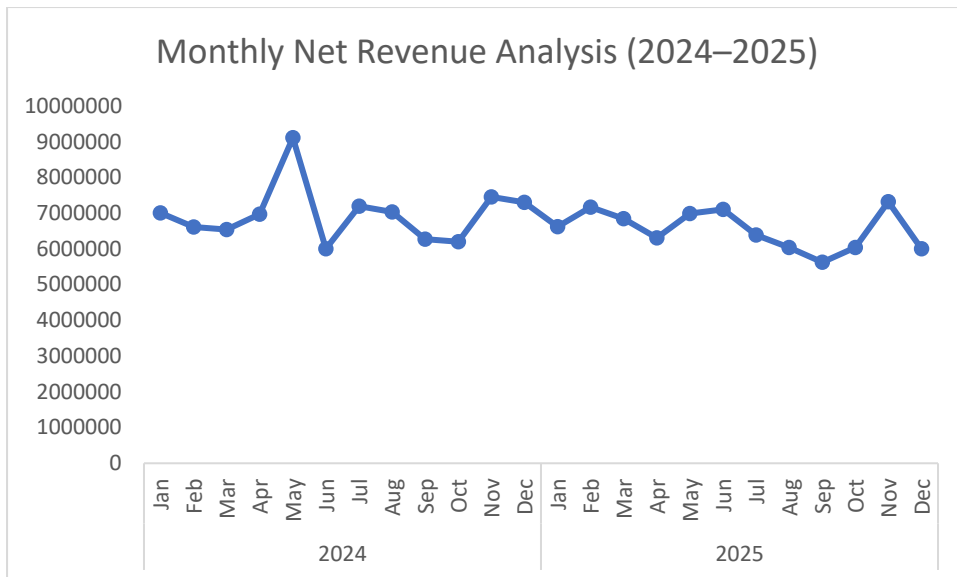
Recommendation

Maintain the current **low-discount pricing approach** unless deeper discounts are proven to generate incremental demand. Focus instead on **bundling, cross-selling and premium product positioning** to increase order value while protecting margins.

Revenue Analysis

Monthly Net Revenue

Year	Month	Net Revenue
2024	January	₹7.00M
2024	February	₹6.61M
2024	March	₹6.54M
2024	April	₹6.97M
2024	May	₹9.11M
2024	June	₹6.01M
2024	July	₹7.19M
2024	August	₹7.03M
2024	September	₹6.27M
2024	October	₹6.20M
2024	November	₹7.46M
2024	December	₹7.31M
2025	January	₹6.63M
2025	February	₹7.17M
2025	March	₹6.85M
2025	April	₹6.31M
2025	May	₹6.99M
2025	June	₹7.11M
2025	July	₹6.39M
2025	August	₹6.04M
2025	September	₹5.63M
2025	October	₹6.04M
2025	November	₹7.32M
2025	December	₹6.00M



Insight

Delivered net revenue totaled approximately **₹83.70M in 2024**, compared with **₹78.48M in 2025**, representing a **6.24% year-over-year decline**.

Revenue also shows noticeable monthly fluctuations. **May 2024 recorded the highest monthly revenue at ₹9.11M**, while **September 2025 was the lowest at ₹5.63M**. In 2025, revenue recovered to **₹7.32M in November** but fell again to **₹6.00M in December**.

The pattern indicates that revenue performance is **not consistently growing**, with several periods of decline and volatility.

Recommendation

Management should investigate the drivers behind the **6.24% annual revenue decline**, particularly the weaker months in 2025. The business should compare **product/category mix, customer activity, order volume, pricing and regional performance** between 2024 and 2025.

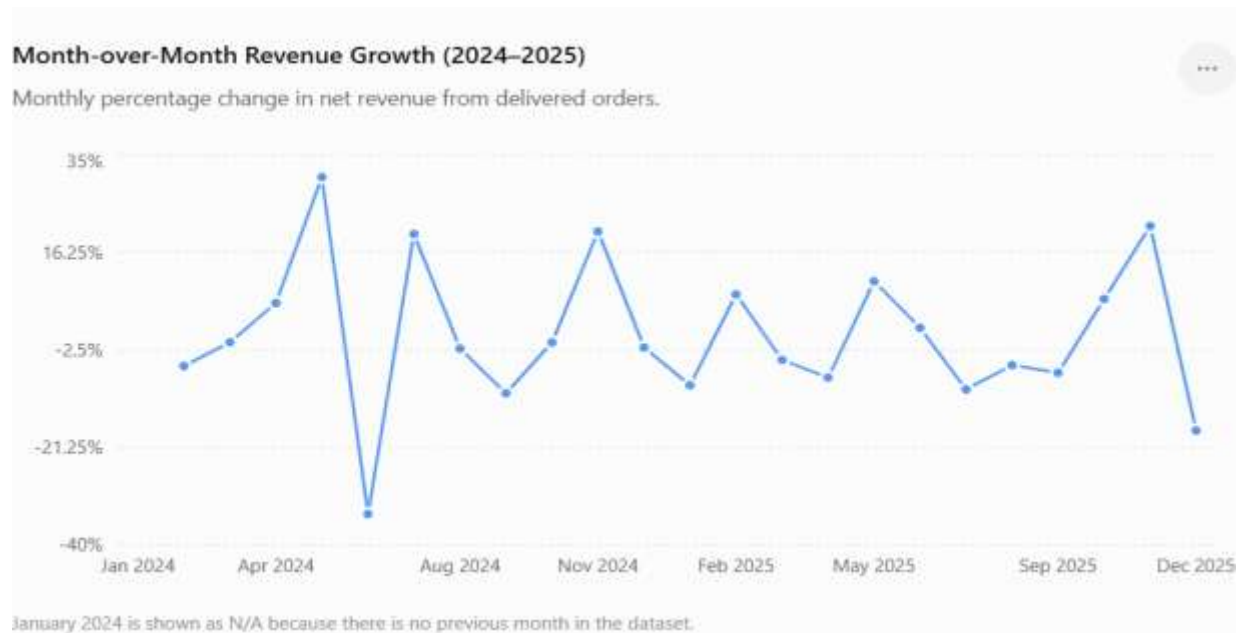
The strong **₹9.11M May 2024 performance** can also be investigated to identify successful products, categories or campaigns that could potentially be replicated during weaker periods such as **September 2025 (₹5.63M)**.

Month-over-Month Revenue Growth

Business Question

Calculate the month-over-month revenue growth percentage for Delivered orders.

Year	Month	Net Revenue	MoM Growth	Growth %
2024	January	₹7.00M	N/A	N/A
2024	February	₹6.61M	-₹0.39M	-5.61%
2024	March	₹6.54M	-₹0.07M	-1.04%
2024	April	₹6.97M	+₹0.43M	+6.51%
2024	May	₹9.11M	+₹2.14M	+30.75%
2024	June	₹6.01M	-₹3.11M	-34.09%
2024	July	₹7.19M	+₹1.19M	+19.79%
2024	August	₹7.03M	-₹0.16M	-2.25%
2024	September	₹6.27M	-₹0.76M	-10.85%
2024	October	₹6.20M	-₹0.07M	-1.07%
2024	November	₹7.46M	+₹1.26M	+20.28%
2024	December	₹7.31M	-₹0.15M	-2.06%
2025	January	₹6.63M	-₹0.68M	-9.31%
2025	February	₹7.17M	+₹0.54M	+8.18%
2025	March	₹6.85M	-₹0.32M	-4.43%
2025	April	₹6.31M	-₹0.54M	-7.84%
2025	May	₹6.99M	+₹0.68M	+10.70%
2025	June	₹7.11M	+₹0.12M	+1.74%
2025	July	₹6.39M	-₹0.72M	-10.12%
2025	August	₹6.04M	-₹0.35M	-5.44%
2025	September	₹5.63M	-₹0.42M	-6.90%
2025	October	₹6.04M	+₹0.41M	+7.30%
2025	November	₹7.32M	+₹1.29M	+21.32%
2025	December	₹6.00M	-₹1.32M	-18.03%



Insight

The revenue trend shows **high month-to-month volatility rather than consistent growth**.

- The strongest positive movement occurred in **May 2024 (+30.75%)**.
- This was followed by the sharpest decline in **June 2024 (-34.09%)**, indicating a major reversal in monthly revenue performance.
- In 2025, revenue declined for **three consecutive months from July through September**, reaching **-6.90% in September**.
- Revenue recovered strongly in **November 2025 (+21.32%)**, but this was followed by another significant decline of **-18.03% in December 2025**.
- The repeated sharp increases followed by declines suggest that revenue performance may be influenced by **seasonality, promotional activity, product mix, customer demand, or changes in order volume**.

Business Recommendation

The business should investigate the causes behind the **large month-to-month swings**, particularly the **May–June 2024 reversal** and the **November–December 2025 reversal**.

Management should compare monthly **order volume, category performance, discount levels, customer activity, and regional revenue** to identify the drivers of these fluctuations.

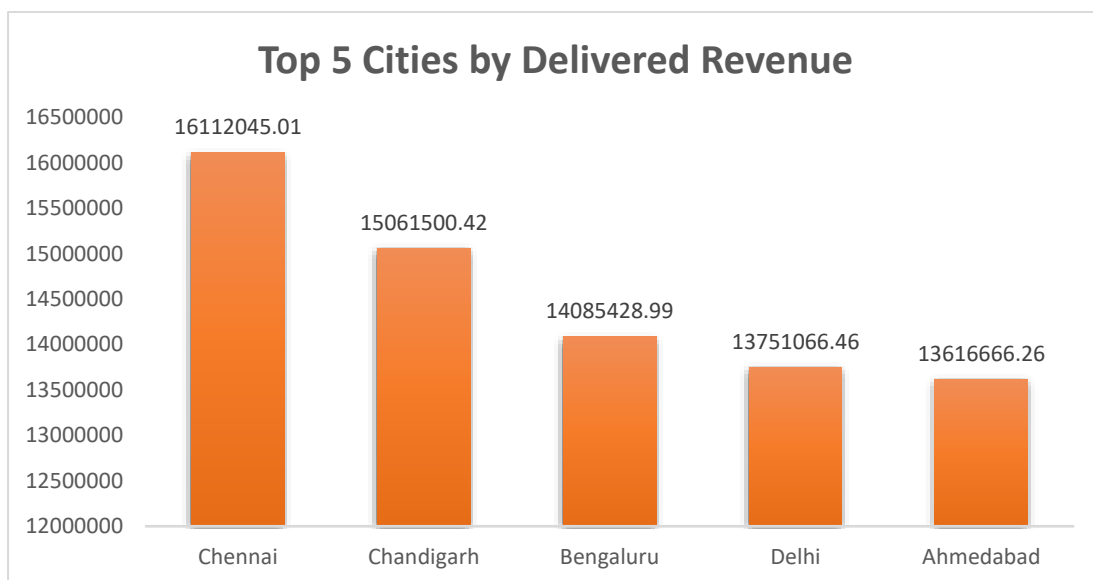
The strong November performance could also be investigated to determine which products, categories, customers, or campaigns contributed to the **+21.32% increase**, and whether those successful factors can be replicated while reducing the sharp declines seen in the following months.

Top 5 Cities by Delivered Revenue

Business Question

Rank cities by delivered revenue and calculate each top city's contribution to total delivered revenue.

Rank	City	Net Revenue	Revenue Contribution
1	Chennai	₹16.11M	9.93%
2	Chandigarh	₹15.06M	9.29%
3	Bengaluru	₹14.09M	8.68%
4	Delhi	₹13.75M	8.48%
5	Ahmedabad	₹13.62M	8.40%



Insight

Chennai is the leading city, generating approximately **₹16.11M** in delivered net revenue and contributing **9.93%** of total delivered revenue. Chandigarh follows with ₹15.06M (9.29%), while Bengaluru, Delhi, and Ahmedabad contribute 8.68%, 8.48%, and 8.40%, respectively.

The relatively close revenue levels among the five cities indicate that delivered revenue is **distributed across multiple major markets rather than being dominated by a single city**.

Business Recommendation

The business should **protect and strengthen performance in the top-performing cities**, particularly Chennai, while investigating the factors driving strong performance in Chandigarh, Bengaluru, Delhi, and Ahmedabad.

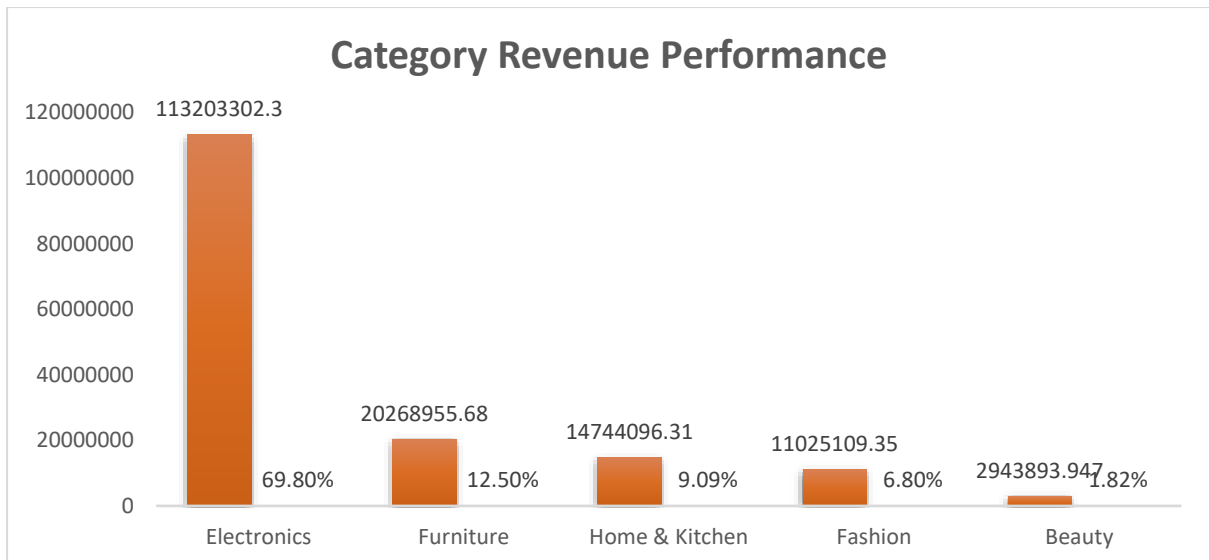
At the same time, management should identify opportunities in lower-performing cities to determine whether targeted **marketing campaigns, product availability, pricing strategies, or customer acquisition initiatives** can increase their contribution.

Category Revenue Performance

Business Question:

Calculate revenue by category, rank categories by revenue, and determine each category's contribution to total delivered revenue.

Rank	Category	Net Revenue	Revenue Contribution
1	Electronics	₹113.20M	69.80%
2	Furniture	₹20.27M	12.50%
3	Home & Kitchen	₹14.74M	9.09%
4	Fashion	₹11.03M	6.80%
5	Beauty	₹2.94M	1.82%



Key Insight

Electronics is the dominant revenue category, generating ₹113.20M and contributing 69.80% of total delivered revenue. Furniture ranked second with ₹20.27M (12.50%), followed by Home & Kitchen at ₹14.74M (9.09%), Fashion at ₹11.03M (6.80%), and Beauty at ₹2.94M (1.82%). The results show a **high concentration of revenue in Electronics**, with the category generating more than two-thirds of total delivered revenue.

Business Recommendation

The business should **protect and strengthen the Electronics category** because of its critical contribution to overall revenue, while reducing dependency on a single category by developing **Furniture, Home & Kitchen, and Fashion** through targeted promotions, product expansion, and cross-selling strategies. The relatively low contribution from **Beauty (1.82%)** should also be investigated to determine whether product demand, assortment, pricing, or customer engagement is limiting its performance.

Top 10 Customers by Total Delivered Revenue

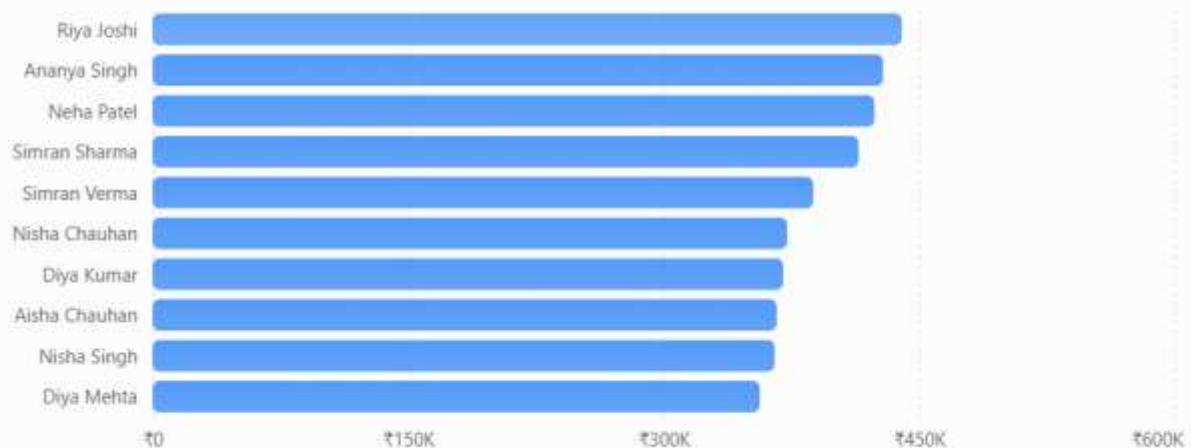
Business Question:

Identify the top 10 customers based on total delivered revenue.

Rank	Customer ID	Customer Name	Net Revenue
1	C5849	Riya Joshi	₹438,716.08
2	C5771	Ananya Singh	₹427,787.00
3	C2474	Neha Patel	₹422,693.87
4	C2180	Simran Sharma	₹413,389.10
5	C4419	Simran Verma	₹386,769.74
6	C1354	Nisha Chauhan	₹371,458.53
7	C5088	Diya Kumar	₹369,113.30
8	C5998	Aisha Chauhan	₹365,364.66
9	C4645	Nisha Singh	₹364,043.04
10	C2246	Diya Mehta	₹355,351.29

Top 10 Customers by Delivered Revenue

Top customers ranked by total net revenue generated from delivered orders.



Key Insight

- **Riya Joshi** is the highest-value customer, generating **₹438.72K** in delivered revenue.
- The second-highest customer, **Ananya Singh**, generated **₹427.79K**, while the remaining customers range from **₹355.35K to ₹422.69K**.
- The relatively narrow gap between the highest and lowest customers in the top 10 indicates that **high-value revenue is distributed across several customers rather than being concentrated in one individual customer**.
- The top 10 customers represent an important **high-value customer segment** that can contribute significantly to revenue stability.

Business Recommendation

- The business should **prioritize retention of these high-value customers** through personalized offers, loyalty benefits, and targeted engagement.
- Analyze their **purchase frequency, preferred categories, average order value, and buying patterns** to understand what drives their high revenue contribution.
- Use these customers as a benchmark to identify **similar high-potential customers** who could be converted into the next tier of high-value customers.

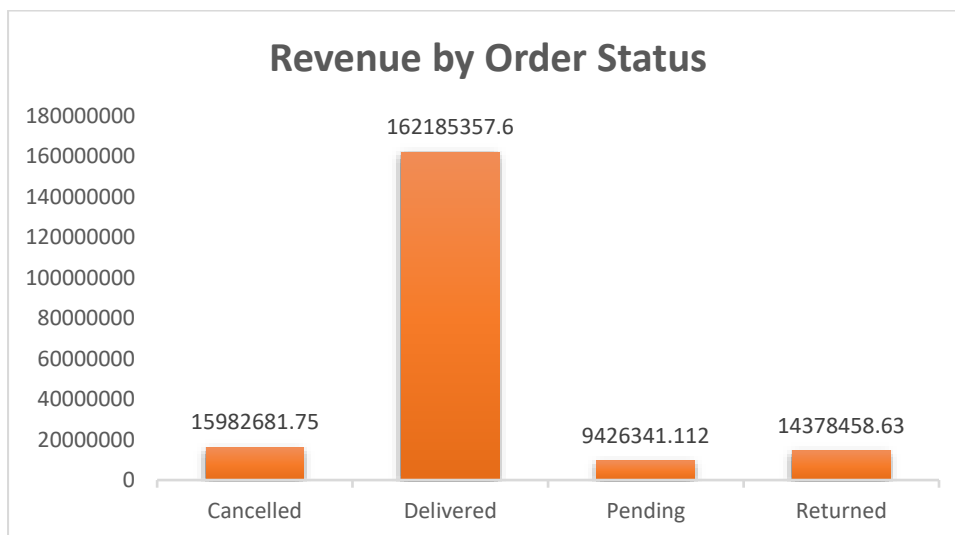
ORDERS & BUSINESS PERFORMANCE

Delivered, Returned, Cancelled & Pending Revenue Performance

Business Question:

Calculate revenue associated with Delivered, Returned, Cancelled, and Pending orders.

Order Status	Revenue	% of Total
Delivered	₹162.19M	80.30%
Cancelled	₹15.98M	7.91%
Returned	₹14.38M	7.12%
Pending	₹9.43M	4.67%
Total	₹201.97M	100.00%



Key Insight

- **Delivered orders generated ₹162.19M**, representing approximately **80.30% of total revenue**, making them the primary source of business revenue.
- **Cancelled orders account for ₹15.98M (7.91%)**, slightly higher than **Returned orders at ₹14.38M (7.12%)**.
- **Pending orders represent ₹9.43M (4.67%)**, indicating a meaningful amount of revenue that has not yet converted into completed orders.
- Combined, **Cancelled, Returned, and Pending orders represent approximately 19.70% of total revenue**, highlighting a significant portion of revenue associated with orders that are not currently delivered.

Business Recommendation

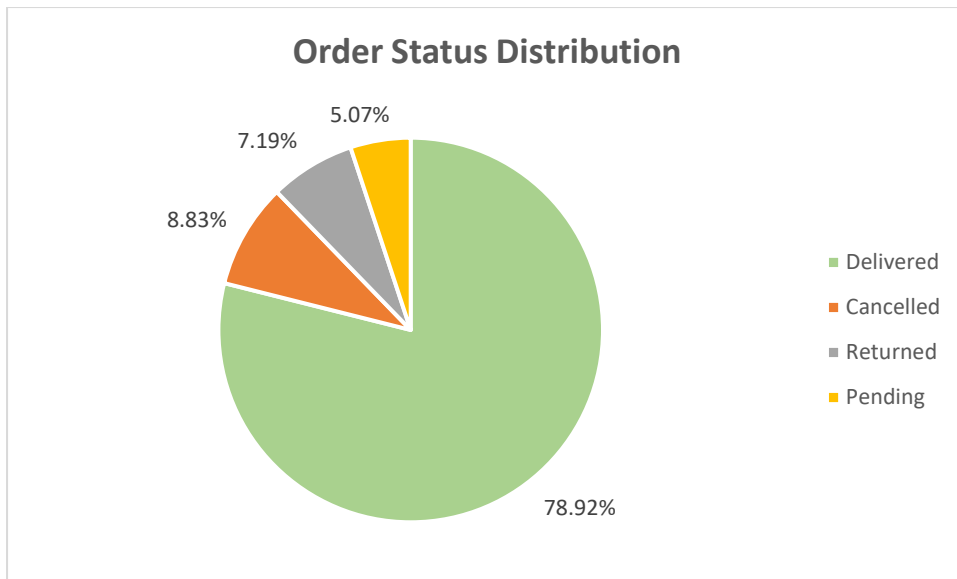
- The business should **prioritize reducing cancellations and returns**, as together they represent approximately **₹30.36M** in revenue associated with unsuccessful or reversed order outcomes.
- Investigate the key drivers of cancellations and returns, such as **product availability, pricing, delivery issues, product quality, and customer expectations**.
- Pending orders should be monitored closely and converted into successful deliveries where possible, since **₹9.43M** is currently tied to pending orders.
- Maintaining the strong **80.30% delivered-revenue share** while reducing non-delivered order outcomes could improve overall revenue realization and operational efficiency.

Order Status Performance

Business Question:

Compare Delivered, Cancelled, Returned, and Pending orders by count and percentage.

Rank	Order Status	Order Count	Order Percentage
1	Delivered	7,831	78.92%
2	Cancelled	876	8.83%
3	Returned	713	7.19%
4	Pending	503	5.07%
Total		9,923	100.01%*



Key Insight

- **Delivered orders dominate the dataset**, with **7,831 orders (78.92%)**, indicating that the majority of placed orders successfully reached the delivered stage.
- **Cancelled orders account for 876 orders (8.83%)**, making cancellations the largest non-delivered category.
- **Returned orders represent 713 orders (7.19%)**, while **503 orders (5.07%)** remain pending.
- Combined, **Cancelled, Returned, and Pending orders account for 2,092 orders**, representing approximately **21.09% of all orders**, indicating a meaningful opportunity to improve overall order completion.

Business Recommendation

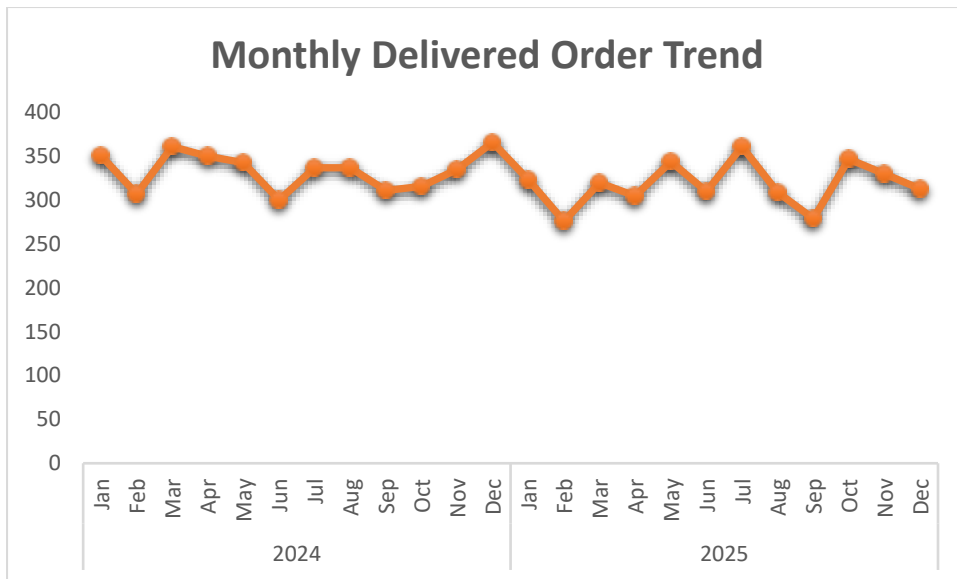
- The business should **prioritize reducing cancellations and returns**, as these two categories together account for **1,589 orders (16.02%)**.
- Investigate the major causes of cancellations and returns, particularly **delivery issues, product availability, pricing, product quality, and customer expectations**.
- Pending orders should be actively monitored to improve conversion into delivered orders and prevent them from becoming cancellations.
- Maintaining the current **78.92% delivery rate** while reducing non-delivered orders could improve overall operational efficiency and customer experience.

Monthly Order Trend

Business Question:

Calculate the monthly number of delivered orders and identify the month with the highest order volume.

Year	Month	Month Name	Delivered Orders
2024	1	January	351
2024	2	February	307
2024	3	March	361
2024	4	April	350
2024	5	May	343
2024	6	June	300
2024	7	July	337
2024	8	August	337
2024	9	September	311
2024	10	October	316
2024	11	November	335
2024	12	December	366
2025	1	January	323
2025	2	February	276
2025	3	March	320
2025	4	April	305
2025	5	May	344
2025	6	June	310
2025	7	July	361
2025	8	August	309
2025	9	September	279
2025	10	October	347
2025	11	November	330
2025	12	December	313



💡 **Key Insight**

- **December 2024 recorded the highest delivered order volume, with 366 orders**, followed closely by March 2024 and July 2025 with **361 orders each**.
- Order volume fluctuated throughout the two-year period, with **2024 generally showing stronger monthly performance** than several corresponding months in 2025
- The lowest order volume occurred in **February 2025 with 276 delivered orders**, followed by September 2025 with 279 orders.
- The pattern indicates **noticeable month-to-month fluctuations**, suggesting that customer demand is not consistent throughout the year.

🎯 **Business Recommendation**

- The business should **investigate the factors behind high-performing months**, particularly December 2024, March 2024, and July 2025, to identify successful promotional, seasonal, or product-related drivers.
- Similar strategies can be tested during historically weaker periods such as **February and September** to improve order volume.
- Management should use monthly order trends for **inventory planning, marketing campaigns, staffing, and promotional scheduling** to better align resources with expected demand.

Number of New Customers Acquired Each Month

Year	Month	New Customers
2024	January	337
2024	February	287
2024	March	310
2024	April	282
2024	May	247
2024	June	209
2024	July	224
2024	August	191
2024	September	177
2024	October	179
2024	November	163
2024	December	164
2025	January	143
2025	February	106
2025	March	124
2025	April	117
2025	May	117
2025	June	105
2025	July	113
2025	August	87
2025	September	69
2025	October	85
2025	November	85
2025	December	67



💡 Key Insights

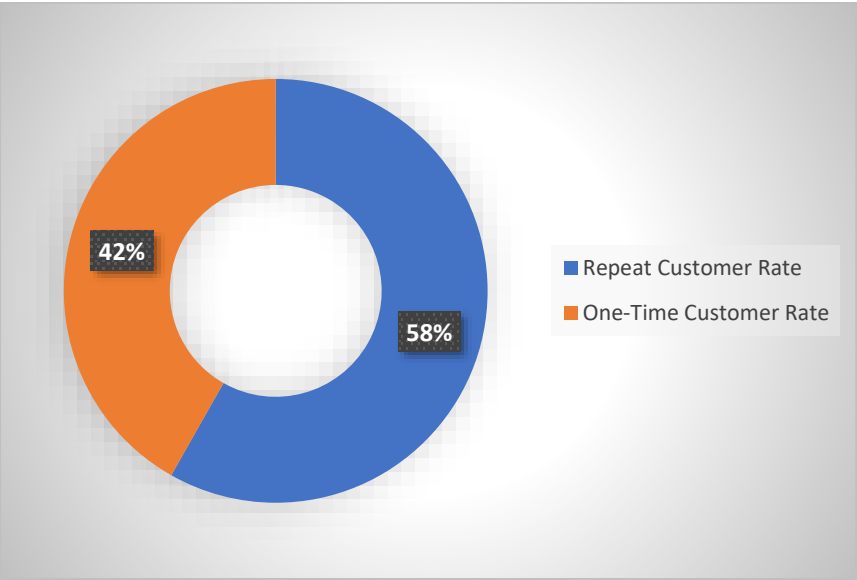
- **New customer acquisition declined substantially over the two-year period, falling from 337 customers in January 2024 to 67 in December 2025.**
- The highest acquisition occurred in **January 2024 with 337 new customers**, followed by **March 2024 with 310**.
- Acquisition remained relatively stronger during the first half of 2024 but **declined progressively toward the end of 2024**.
- **2025 consistently recorded lower new-customer acquisition** than the corresponding periods of 2024.
- The lowest acquisition occurred in **December 2025 with only 67 new customers**, indicating a significant slowdown in customer acquisition.

🎯 Business Recommendations

- **Investigate the decline in new-customer acquisition**, particularly the substantial reduction observed throughout 2025.
- Strengthen **customer acquisition campaigns** through targeted digital marketing, referral programs, first-order offers, and new-customer promotions.
- Analyze the channels and campaigns that generated strong acquisition during early 2024 and **replicate the most effective strategies**.
- Focus additional acquisition efforts during weaker periods such as **August–December 2025**, where monthly new-customer counts remained relatively low.
- Monitor **new-customer acquisition monthly** as a core KPI to identify whether marketing initiatives are reversing the downward trend.

Repeat Customer Analysis

Metric	Result
Total Customers	3,988
Repeat Customers	2,321
Repeat Customer Rate	58.20%
One-Time Customers	1,667
One-Time Customer Rate	41.80%



💡 Key Insights

- 58.20% of customers placed more than one delivered order, indicating a relatively strong repeat-purchase base.
- However, 41.80% of customers placed only one delivered order, representing a significant opportunity to improve customer retention and repeat purchases.

🎯 Business Recommendations

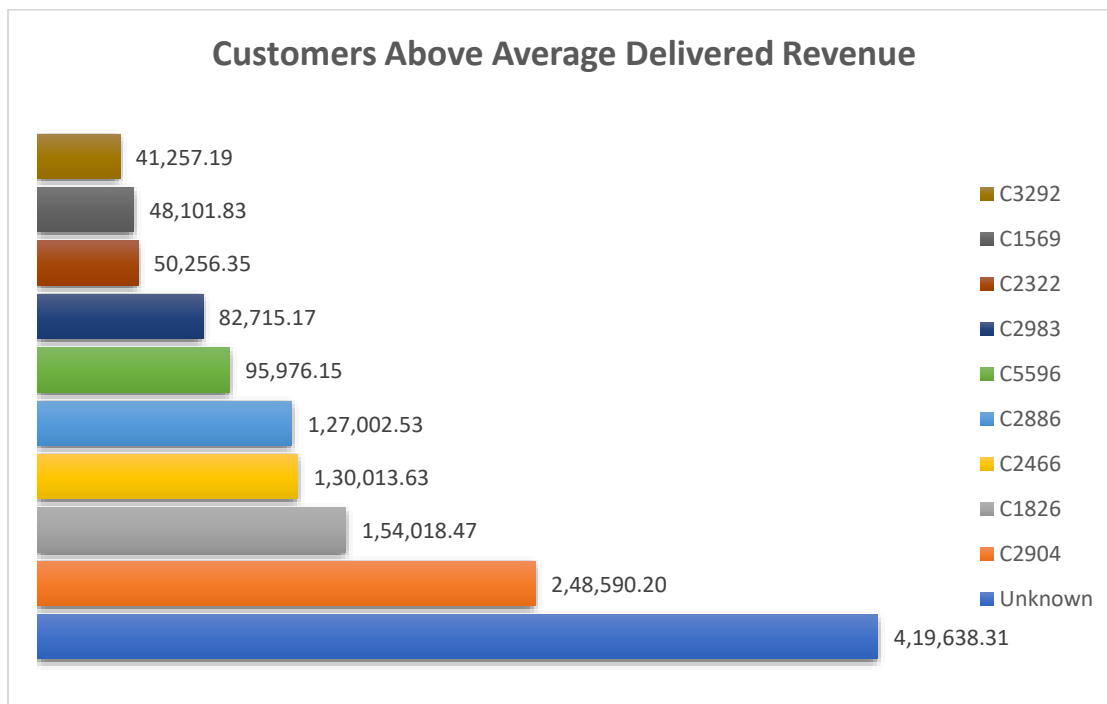
- Strengthen retention through personalized offers, loyalty rewards, and targeted remarketing campaigns to convert one-time customers into repeat buyers.

Customers Above Average Delivered Revenue

Customers with Total Delivered Revenue Above Average

Rank	Customer ID	Net Revenue	Revenue Contribution
1	Unknown	419,638.31	0.26%
2	C2904	248,590.20	0.15%
3	C1826	154,018.47	0.09%
4	C2466	130,013.63	0.08%
5	C2886	127,002.53	0.08%
6	C5596	95,976.15	0.06%
7	C2983	82,715.17	0.05%
8	C2322	50,256.35	0.03%
9	C1569	48,101.83	0.03%
10	C3292	41,257.19	0.03%

Note: Unknown represents customer records where the Customer ID was missing in the original data and was retained as a placeholder during data cleaning.



Key Insights

- **Unknown** customer records generated the highest revenue at **248? Wait** — your actual value is **419,638.31**, making it the highest in this result.
- **C2904** generated the second-highest revenue at **248,590.20**.
- **C1826** ranked third with **154,018.47** in delivered revenue.
- The revenue contribution of each individual customer is relatively small, with the highest identifiable contribution being **0.15%** for C2904.
- The presence of **Unknown** as the highest-revenue customer highlights the importance of improving customer-ID completeness and data quality.

Business Recommendations

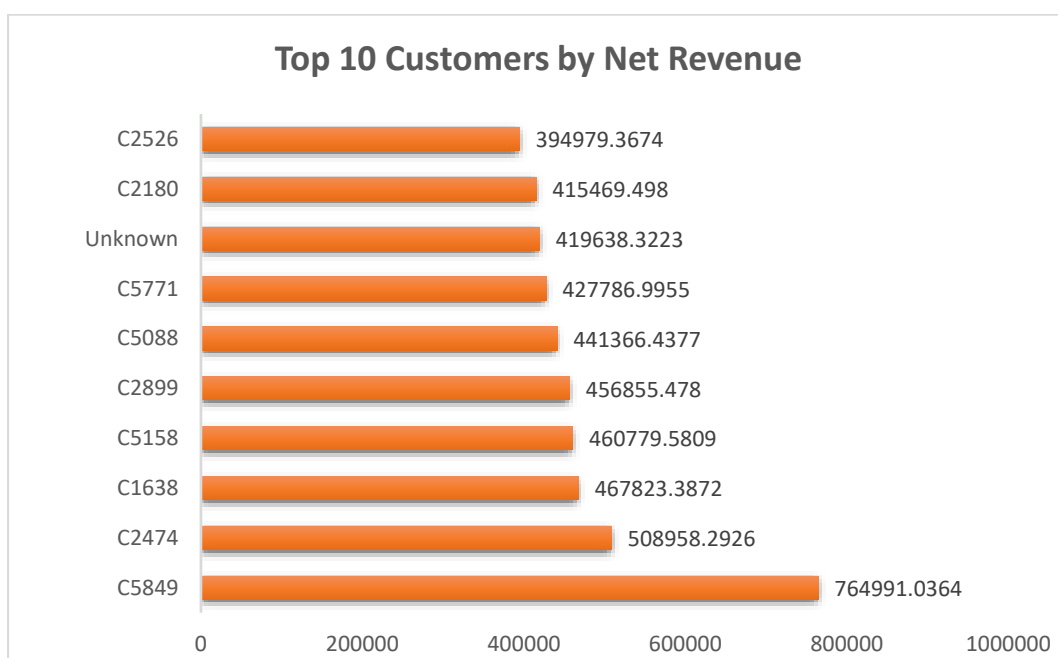
- **Prioritize high-value customers:** Develop loyalty benefits and personalized offers for customers generating above-average revenue.
- **Improve customer identification:** Investigate the Unknown customer records because they represent a significant revenue value without a usable customer identifier.
- **Increase retention:** Monitor high-value customers and proactively engage them when their purchase frequency starts declining.
- **Personalize marketing:** Use purchase history and product preferences to provide targeted recommendations and relevant offers.

Customer-Level Purchase Behavior

Top 10 Customers by Purchase Behavior

Rank	Customer ID	Total Orders	Total Quantity Purchased	Net Revenue	AOV
1	C5849	4	16	764,991.04	191,247.76
2	C2474	3	12	508,958.29	169,652.76
3	C1638	4	11	467,823.39	116,955.85
4	C5158	3	9	460,779.58	153,593.19
5	C2899	2	7	456,855.48	228,427.74
6	C5088	4	9	441,366.43	110,341.61
7	C5771	1	6	427,787.00	427,787.00
8	Unknown	16	33	419,638.31	26,227.39
9	C2180	2	7	415,469.50	207,734.75
10	C2526	3	9	394,979.37	131,659.79

Note: Unknown represents customer records where the Customer ID was missing in the original data and was retained as a placeholder during data cleaning.



Key Insights

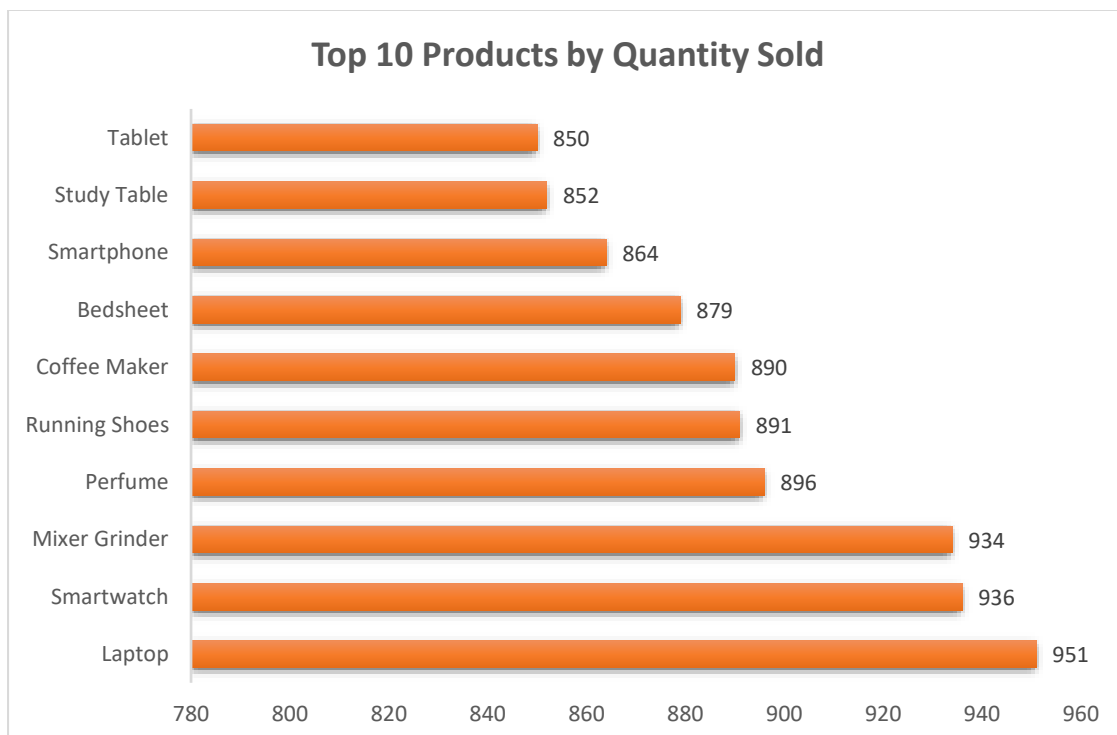
- **C5849** is the highest-value customer, generating **764,991.04** in delivered net revenue across 4 orders.
- **C2899** has the **highest AOV among identified customers** at **228,427.74**, despite placing only 2 orders.
- **C5771** generated **427,787.00 from a single order**, resulting in the highest overall AOV of **427,787.00**.
- **C5088** and **C1638** generated high revenue through relatively frequent purchasing, with 4 delivered orders each.
- The **Unknown** customer record has 16 orders and 33 units but a relatively low AOV of **26,227.39**, highlighting a customer-identification/data-quality issue.

Business Recommendations

- **Prioritize high-value customers:** Develop loyalty benefits and personalized offers for customers generating the highest revenue.
- **Increase retention:** Monitor high-value customers and proactively engage them when purchase frequency declines.
- **Encourage repeat purchases:** Customers with high AOV but fewer orders, such as C5771 and C2899, are strong opportunities for targeted retention campaigns.
- **Personalize marketing:** Use purchase history and customer behavior to recommend relevant products and bundles.
- **Improve customer identification:** Investigate the Unknown customer records because 16 orders are associated with an unidentified customer ID.

Top 10 Products by Quantity Sold

Rank	Product	Total Quantity Sold	Net Revenue
1	Laptop	951	61,649,291.70
2	Smartwatch	936	6,032,951.78
3	Mixer Grinder	934	3,920,550.65
4	Perfume	896	2,494,747.09
5	Running Shoes	891	3,754,995.04
6	Coffee Maker	890	3,988,324.63
7	Bedsheet	879	1,580,681.12
8	Smartphone	864	24,158,694.19
9	Study Table	852	6,409,659.99
10	Tablet	850	18,706,481.55



Key Insights

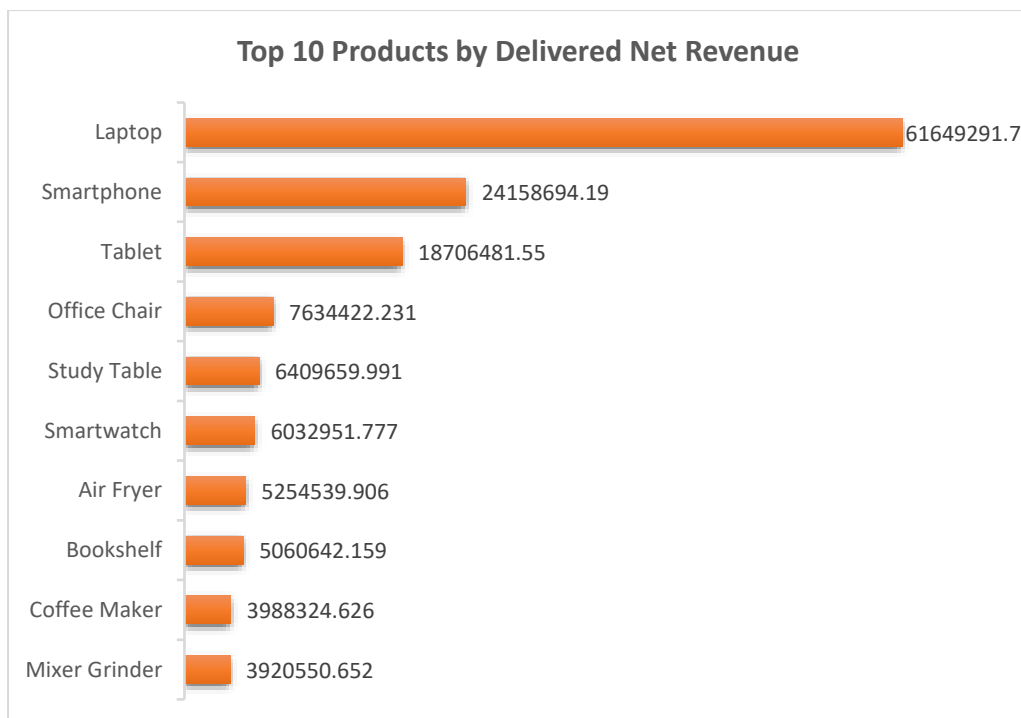
- **Laptop** recorded the highest sales volume with **951 units sold** and generated **61.65M in net revenue**.
- **Smartwatch** ranked second in quantity sold with **936 units**, while generating **6.03M in revenue**.
- **Smartphone** and **Tablet** sold fewer units than several products but generated substantially higher revenue, indicating higher-value products.
- **Bedsheet** sold **879 units** but generated only **1.58M**, making it the lowest-revenue product among the top 10 by volume.
- The results show that **high sales volume does not necessarily translate into high revenue**, highlighting differences in product pricing.

Business Recommendations

- **Prioritize high-volume products:** Maintain sufficient inventory for products such as Laptops, Smartwatches, and Mixer Grinders to avoid stockouts.
- **Focus on high-value products:** Products such as Laptops, Smartphones, and Tablets should receive strategic attention because they generate substantial revenue.
- **Review low-revenue/high-volume products:** Analyze pricing, discounts, and margins for products such as Bedsheets and Perfumes to determine whether profitability can be improved.
- **Use product-specific strategies:** Combine sales volume and revenue when making inventory, pricing, and promotional decisions rather than relying on quantity sold alone.

Top 10 Products by Delivered Net Revenue

Rank	Product	Net Revenue	Revenue Contribution
1	Laptop	61,649,291.77	38.01%
2	Smartphone	24,158,694.26	14.90%
3	Tablet	18,706,481.50	11.53%
4	Office Chair	7,634,422.23	4.71%
5	Study Table	6,409,660.02	3.95%
6	Smartwatch	6,032,951.73	3.72%
7	Air Fryer	5,254,539.85	3.24%
8	Bookshelf	5,060,642.16	3.12%
9	Coffee Maker	3,988,324.57	2.46%
10	Mixer Grinder	3,920,550.66	2.42%



Key Insights

- **Laptop is the strongest revenue-generating product**, contributing **38.01% of total delivered revenue**.
- **Smartphone and Tablet** rank second and third, contributing **14.90% and 11.53%**, respectively.
- The top three products collectively contribute approximately **64.44% of delivered revenue**, showing a strong concentration of revenue among major products.
- **Office Chair** is the fourth-highest revenue contributor at **4.71%**.
- The results show that a small number of products are responsible for a substantial share of overall revenue.

Business Recommendations

- **Prioritize high-revenue products:** Maintain sufficient inventory and availability for Laptops, Smartphones, and Tablets to protect the major revenue streams.
- **Reduce revenue concentration risk:** Continue developing other promising products so the business is not overly dependent on a small number of products.
- **Use targeted promotions:** Promote mid-tier products such as Office Chairs, Study Tables, and Smartwatches to increase their revenue contribution.
- **Monitor top-product performance:** Track pricing, discounts, inventory, and demand for the top three products regularly because their performance has a major impact on overall revenue.

Top 3 Products by Net Revenue in Each Category

Category	Product	Net Revenue	Rank
Beauty	Perfume	2,494,747.09	1
Beauty	Face Wash	449,146.79	2
Electronics	Laptop	61,649,291.77	1
Electronics	Smartphone	24,158,694.26	2
Electronics	Tablet	18,706,481.50	3
Fashion	Running Shoes	3,754,994.98	1
Fashion	Jacket	2,993,434.54	2
Fashion	Jeans	1,858,587.68	3
Furniture	Office Chair	7,634,422.23	1
Furniture	Study Table	6,409,660.02	2
Furniture	Bookshelf	5,060,642.16	3
Home & Kitchen	Air Fryer	5,254,539.85	1
Home & Kitchen	Coffee Maker	3,988,324.57	2
Home & Kitchen	Mixer Grinder	3,920,550.66	3

Key insight

Electronics is strongly dominated by Laptop revenue, with Laptop generating approximately **₹61.65M**, substantially higher than the other top products. Smartphone and Tablet follow at approximately **₹24.16M** and **₹18.71M** respectively.

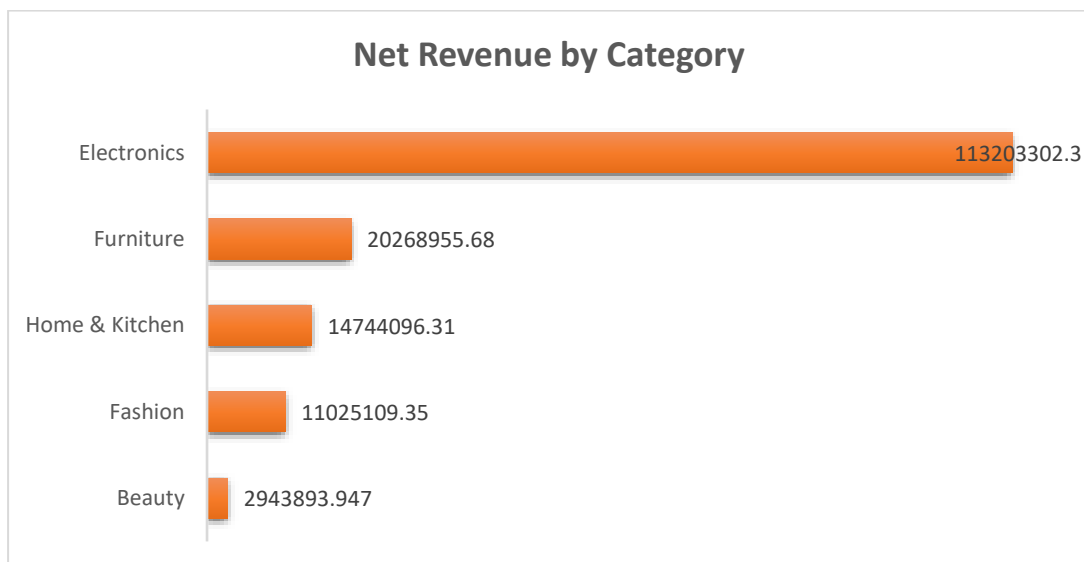
Other categories show more balanced product performance. In Furniture, for example, Office Chair, Study Table, and Bookshelf all contribute substantial revenue, while Fashion and Home & Kitchen also have multiple products generating meaningful revenue.

Business recommendation

Prioritize inventory availability and promotional investment for the strongest products, particularly Laptop, Smartphone, and Tablet. At the same time, investigate why lower-ranked products contribute less revenue and evaluate opportunities such as bundling, cross-selling, targeted discounts, or improved product promotion to strengthen the weaker products.

Category Performance — Orders, Quantity, Revenue, AOV & Discount

Category	Total Orders	Quantity Sold	Avg. Discount (%)	Net Revenue	AOV	Revenue Rank
Electronics	1,978	4,358	0.12	113,203,302.29	57,231.19	1
Furniture	1,564	3,381	0.12	20,268,955.68	12,959.69	2
Home & Kitchen	1,578	3,514	0.12	14,744,096.15	9,343.53	3
Fashion	1,910	4,244	0.13	11,025,109.34	5,772.31	4
Beauty	801	1,717	0.12	2,943,893.88	3,675.27	5



Key Insights

- **Electronics is the strongest-performing category**, generating ₹113.20M in delivered revenue and ranking #1.
- Electronics also has the **highest AOV at ₹57,231.19**, significantly higher than every other category.
- **Fashion has the highest average discount at 0.13%**, while the other categories average around 0.12%.
- **Beauty has the lowest revenue (₹2.94M), lowest order count (801), and lowest AOV (₹3,675.27).**
- Electronics generates substantially more revenue than the other categories, indicating a strong concentration of revenue in this category.

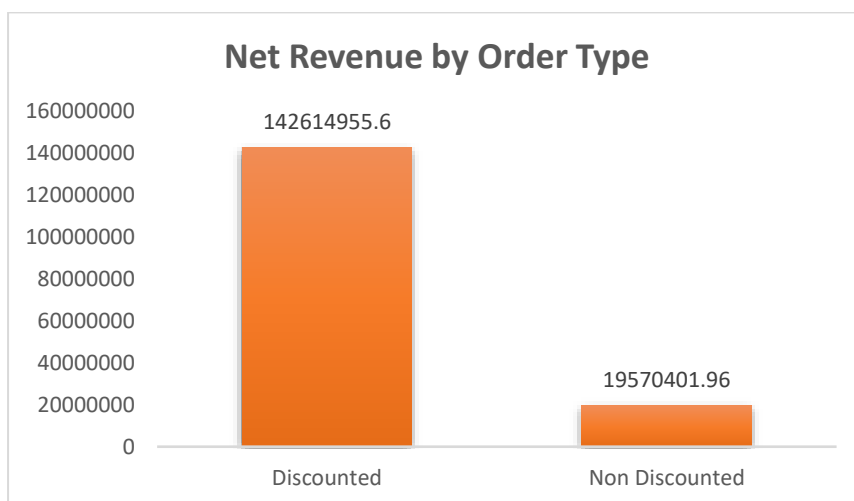
Business Recommendations

- **Prioritize Electronics:** Maintain inventory availability and marketing focus on high-value electronics products because the category is the major revenue driver.
- **Increase investment in Furniture and Home & Kitchen:** These categories generate meaningful revenue but have considerably lower AOV than Electronics, creating an opportunity for upselling and cross-selling.
- **Improve Fashion profitability:** Since Fashion has the highest average discount, review discount strategies to ensure promotions are contributing to incremental sales without unnecessarily reducing revenue.
- **Grow Beauty:** Use targeted promotions, product bundles, and cross-selling with higher-performing categories to increase order volume and AOV.
- **Monitor category concentration:** Continue tracking Electronics' share of revenue so the business can maintain its strongest category while gradually diversifying revenue across other categories.

DISCOUNT & PRICING ANALYSIS

Discounted vs Non-Discounted Orders — Revenue & AOV Comparison

Order Type	Total Orders	Net Revenue	AOV
Discounted	6,886	142,614,955.64	20,710.86
Non-Discounted	945	19,570,401.96	20,709.42



Key Insights

- **Discounted orders generated ₹142.61M in net revenue**, making them the dominant revenue source among delivered orders.
- Discounted orders accounted for **6,886 orders**, compared with only **945 non-discounted orders**.
- **AOV is almost identical** between the two groups: ₹20,710.86 for discounted orders versus ₹20,709.42 for non-discounted orders.
- The AOV difference is only **₹1.44**, indicating that discounts did **not materially increase the average order value** in this dataset.
- The much higher revenue from discounted orders is therefore primarily associated with their **much larger order volume**, rather than a higher AOV.

Business Recommendations

- **Continue using discounts strategically**, because discounted orders represent a major portion of delivered revenue.
- **Do not assume that discounts increase AOV**, since the AOV is virtually the same for discounted and non-discounted orders.
- **Evaluate discount effectiveness by product/category** to identify where discounts actually generate incremental sales rather than simply reducing realized revenue.
- **Prioritize targeted discounts over blanket discounts**, focusing promotions on products or customer segments where discounts can increase order frequency or conversion.
- **Monitor margin impact** alongside revenue before increasing discount levels, because high revenue from discounted orders does not automatically mean higher profitability.

Discount Band Performance — Orders, Net Revenue & AOV

Discount Band	Total Orders	Net Revenue	AOV
No Discount	945	19,570,401.96	20,709.42
1–10%	6,886	142,614,955.64	20,710.86



Key Insights

- **1–10% discount orders generated ₹142.61M in net revenue**, substantially higher than the ₹19.57M generated by non-discounted orders.
- The 1–10% discount band accounted for **6,886 delivered orders**, compared with 945 non-discounted orders.
- **AOV remained almost identical** between the two bands: ₹20,710.86 for 1–10% discounts versus ₹20,709.42 for no discount.
- The results indicate that the higher revenue from the 1–10% band is primarily driven by **higher order volume rather than a meaningful increase in AOV**.
- No delivered orders fall into the **11–20%, 21–30%, or above-30% discount bands**, based on this dataset

Business Recommendations

- **Continue evaluating 1–10% discounts as a targeted promotional range**, since this band is associated with the majority of delivered orders and revenue.
- **Avoid assuming that higher discounts will increase AOV**, because even the existing 1–10% discount band produces virtually the same AOV as non-discounted orders.
- **Test higher discount bands carefully** if the business wants to introduce them, monitoring order volume, revenue, and profitability rather than revenue alone
- **Analyze discount effectiveness by category and product** to determine where discounts are actually driving additional demand.
- **Track profit margin alongside net revenue** before expanding discounting, since additional revenue does not necessarily translate into additional profit.

Products with Above-Category-Average Discounts

Category	Product	Product Avg. Discount	Category Avg. Discount
Electronics	Smartphone	0.13%	0.12%
Furniture	LED Lamp	0.13%	0.12%



Key Insights

- **Smartphone** in Electronics has an average discount of **0.13%**, slightly above the Electronics category average of **0.12%**.
- **LED Lamp** in Furniture also has an average discount of **0.13%**, compared with the Furniture category average of **0.12%**.
- Both products are only **0.01 percentage points above** their respective category averages, indicating a **very small discount deviation**.
- Only **two products** meet the above-category-average condition in your result, suggesting that discounting is relatively consistent across most products.

Business Recommendations

- **Monitor Smartphone and LED Lamp discounting** to determine whether the additional discount is generating enough incremental sales to justify it.
- Avoid increasing discounts automatically; first compare **sales volume, revenue, and margin** against other products in the same category.
- Use category averages as a **benchmark** for identifying products with unusually high promotional activity.
- If these products generate strong sales without significantly higher discounts, maintain the current discount strategy rather than increasing discounts further.

DELIVERY AND OPERATIONAL ANALYSIS

Average Delivery Time for Delivered Orders

Metric	Result
Average Delivery Time	3.71 days

Insight:

Delivered orders were completed in an average of **3.71 days**, establishing the overall delivery-time benchmark.

Recommendation:

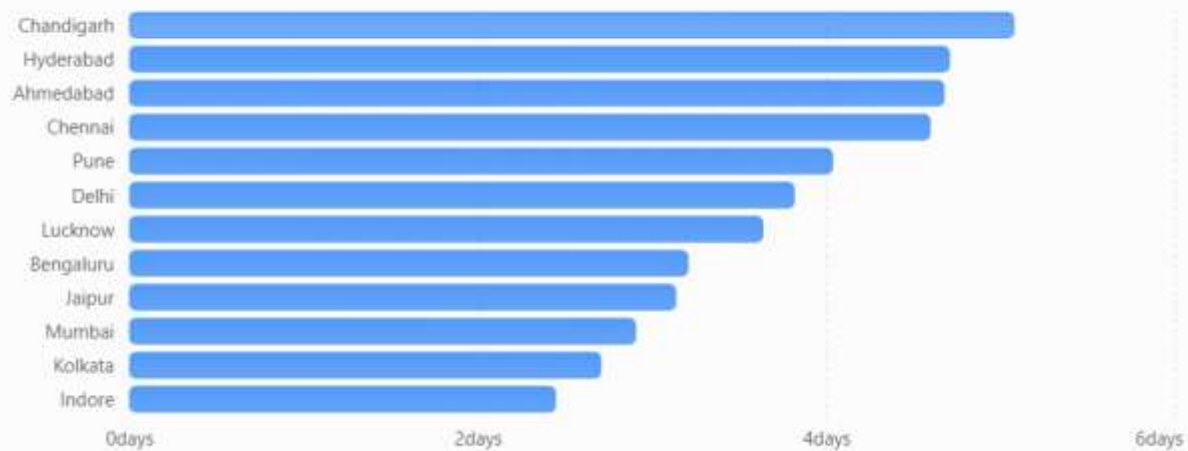
Use **3.71 days as the baseline** and focus operational improvements on cities exceeding this benchmark.

Average Delivery Time by City

City	Delivered Orders	Avg. Delivery Time
Chandigarh	646	5.07 days
Hyderabad	661	4.70 days
Ahmedabad	627	4.67 days
Chennai	671	4.59 days
Pune	648	4.03 days
Delhi	680	3.81 days
Lucknow	659	3.63 days
Bengaluru	650	3.20 days
Jaipur	614	3.13 days
Mumbai	623	2.90 days
Kolkata	648	2.70 days
Indore	647	2.44 days

Average Delivery Time by City

Average delivery time for delivered orders across cities.



Insight

- **Chandigarh** records the highest average delivery time at **5.07 days**, followed by Hyderabad (**4.70 days**) and Ahmedabad (**4.67 days**).
- **Indore** has the fastest average delivery time at **2.44 days**, followed by Kolkata (**2.70 days**) and Mumbai (**2.90 days**).
- **4 cities** — Chandigarh, Hyderabad, Ahmedabad, and Chennai — have average delivery times above **4.5 days**, indicating weaker delivery performance.

Recommendation

- Prioritize **Chandigarh, Hyderabad, Ahmedabad, and Chennai** for operational review and identify the causes of longer delivery times.
- Evaluate **logistics partners, fulfillment processes, and transportation routes** in high-delay cities.
- Use **Indore, Kolkata, and Mumbai** as benchmarks to identify practices that could be replicated in slower-performing cities

Cumulative Revenue Records by Customer

Customer	Date	Daily Revenue	Cumulative Revenue
C5849	2025-05-24	438,716.08	764,991.04
C2474	2024-11-13	422,693.87	508,958.29
C1638	2025-11-04	479.28	467,823.39
C1638	2025-06-30	11,102.72	467,344.11
C5158	2025-08-11	199,573.90	460,779.58
C2899	2025-09-26	248,156.58	456,855.48
C1638	2025-06-24	180,680.38	456,241.39
C5088	2025-12-04	369,113.30	441,366.43
C5771	2025-11-12	427,787.00	427,787.00
Unknown	2025-11-15	8,821.33	419,638.31

Insight

- **C5849** reaches the highest cumulative revenue of **764,991.04**, followed by **C2474** at **508,958.29**.
- **C1638** appears multiple times because cumulative revenue is tracked across different dates, showing the customer's revenue building over time.
- The presence of **Unknown** with **419,638.31** in cumulative revenue highlights the potential business impact of incomplete customer identification.

Recommendation

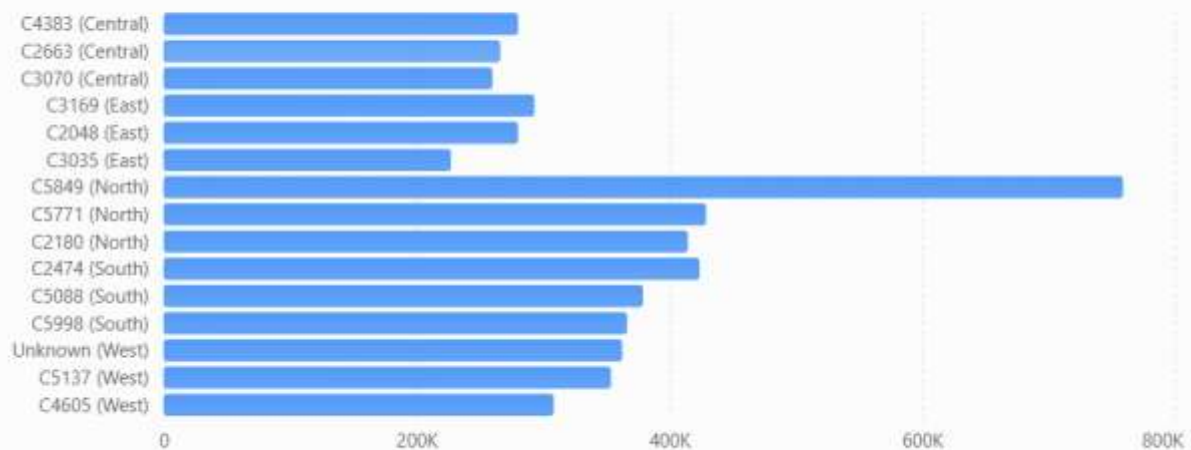
- Prioritize **high-value customers** such as C5849 and C2474 with retention and loyalty strategies.
- Investigate the **Unknown customer records** to improve customer identification and ensure accurate customer-level reporting.
- Use cumulative revenue trends to identify customers whose spending is increasing over time and target them with personalized offers.

Top 3 Customers by Net Revenue in Each Region

Region	Customer	Net Revenue	Rank
Central	C4383	279,118.95	1
Central	C2663	264,972.18	2
Central	C3070	258,854.61	3
East	C3169	292,096.00	1
East	C2048	279,114.53	2
East	C3035	225,978.41	3
North	C5849	757,708.99	1
North	C5771	427,787.00	2
North	C2180	413,389.10	3
South	C2474	422,693.87	1
South	C5088	377,943.88	2
South	C5998	365,364.66	3
West	Unknown*	361,532.78	1
West	C5137	352,672.27	2
West	C4605	307,327.12	3

Top 3 Customers by Region

Net revenue generated by the three highest-revenue customers within each region.



Insights

- **North has the strongest high-value customers**, with C5849 generating **757.7K**, significantly ahead of the top customers in other regions.
- **North shows a wider gap between its top customer and the next two customers**, indicating that C5849 is an especially important high-value customer.
- **South has a relatively strong and balanced top-customer segment**, with all three top customers generating more than **365K** each.
- **Central and East have more closely distributed customer revenue**, suggesting that revenue is less concentrated among a single top customer.
- In **West**, the highest-revenue record belongs to **Unknown Customer ID**, generating **361.5K**. This represents a customer-data quality issue because a significant revenue contribution cannot be attributed to a known customer.
- **C5849 appears as the strongest customer across the regional analysis**, making this customer a potential candidate for VIP/loyalty retention strategies.
- The regional ranking helps identify **high-value customers by geography**, which can support region-specific customer retention and marketing strategies.

Business Recommendations

- Create **VIP/loyalty programs** for the highest-value customers, especially C5849 and other customers generating consistently high revenue.
- Focus retention efforts on the **North region**, where high-value customers contribute particularly large revenue amounts.
- Investigate and resolve **Unknown Customer ID records** so high-value revenue can be properly attributed to customers.
- Use these regional customer rankings to develop **targeted promotions and personalized offers** rather than applying the same strategy across all regions.

Business Questions & Key Findings

1. How is revenue performing over time, and which periods generate the highest sales?

Answer:

Revenue analysis shows that sales performance varies across periods, with certain months and years contributing significantly more revenue than others. The analysis helps identify high-performing periods and supports better planning of promotions, inventory, and sales activities.

2. Which cities, regions, categories, and products contribute the most to revenue?

Answer:

Revenue is highly concentrated in specific products and categories.

Electronics is the strongest-performing category, generating approximately **113.20M** in net revenue. At the product level, **Laptop** is the leading product, contributing approximately **61.65M**, followed by Smartphone and Tablet.

3. What is the overall order performance across delivered, cancelled, returned, and pending orders?

Answer:

Order-status analysis provides an overview of successful and unsuccessful orders and highlights the proportion of delivered, cancelled, returned, and pending transactions. This helps evaluate overall order efficiency and identify potential revenue leakage from unsuccessful orders.

4. How many new and repeat customers does the business acquire, and which customers generate the most revenue?

Answer:

The analysis identified **3,988 customers**, of which **2,321 are repeat customers**, resulting in a **58.20% repeat-customer rate**. High-value customers contribute substantial revenue, with the top customer generating approximately **764.99K** in cumulative revenue.

5. How does customer purchasing behavior vary across the business?

Answer:

Customer-level analysis shows significant variation in **order frequency, quantity purchased, revenue generated, and AOV**. Some customers generate substantial revenue through fewer high-value orders, while others contribute through more frequent purchases, highlighting different customer-value patterns.

6. Which products and categories demonstrate the strongest sales performance?

Answer:

Electronics leads overall category performance, with approximately **4,358 units sold** and **113.20M** in net revenue. At the product level, **Laptop, Smartphone, and Tablet** are the strongest revenue-generating products, while Laptop also leads in quantity sold.

7. How do discounts affect revenue and average order value?

Answer:

Discounted orders generated approximately **142.61M** in net revenue compared with **19.57M** from non-discounted orders. However, AOV was almost identical—**20,710.86** for discounted orders versus **20,709.42** for non-discounted orders—indicating that discounts generated substantially more revenue without materially increasing AOV.

8. How does delivery performance vary across cities and categories?

Answer:

The overall average delivery time is **3.71 days**. Delivery performance varies considerably by city, with **Chandigarh having the highest average delivery time at 5.07 days**, while **Indore has the lowest at 2.44 days**. This indicates meaningful variation in operational efficiency across locations.

9. Which business areas present opportunities to improve revenue, customer engagement, and operational performance?

Answer:

The analysis highlights opportunities in **high-value customer retention, regional delivery optimization, product/category performance, discount strategy, and customer-data quality**. Particular attention should be given to high-performing products and customers while addressing slower delivery locations and **Unknown Customer ID** records.

Insights, Recommendations & Conclusion

Key Insights

1. Revenue is highly concentrated in Electronics

Electronics is the strongest-performing category, generating approximately **113.20M in net revenue**, substantially ahead of the other categories. At the product level, **Laptop** is the leading revenue contributor at approximately **61.65M**, followed by Smartphone and Tablet.

2. High-value customers make a significant contribution

Customer-level analysis identified **3,988 customers**, with **2,321 repeat customers**, resulting in a **58.20% repeat-customer rate**. The highest-value customers generate substantial revenue, indicating the importance of customer retention and targeted engagement.

3. Customer value varies considerably

Customer purchasing behavior differs across order frequency, quantity purchased, revenue, and AOV. Some customers generate high revenue through fewer high-value orders, while others contribute through repeated purchases, indicating different customer-value segments.

4. Discounted orders generate substantially more revenue

Discounted orders generated approximately **142.61M** in net revenue compared with **19.57M** from non-discounted orders. However, AOV remained almost identical at **20,710.86** and **20,709.42**, suggesting that discounts increased sales participation without materially increasing average order value.

5. Revenue performance differs significantly across regions

The regional customer analysis shows that high-value customers are not evenly distributed. **North** contains particularly strong high-value customers, led by C5849 with approximately **757.71K** in revenue.

6. Product performance is concentrated among a few key products

Laptop, Smartphone, and Tablet are the leading revenue-generating products. The top three products within several categories also account for a meaningful share of category revenue, making product-level performance important for inventory and marketing decisions.

7. Delivery performance varies considerably by city

The overall average delivery time for delivered orders is **3.71 days**. However, performance varies substantially across cities, with **Chandigarh averaging 5.07 days**, compared with **Indore at 2.44 days**. This indicates opportunities to improve logistics performance in slower locations

8. Customer data quality can affect business analysis

The presence of **Unknown Customer IDs** among high-revenue records indicates that incomplete customer identification can affect customer-level revenue attribution and segmentation. Improving customer-data completeness would increase the reliability of customer analytics.

Business Recommendations

1. Prioritize high-performing products

Maintain strong inventory availability and targeted marketing for **Laptop, Smartphone, Tablet**, and other high-revenue products to protect the company's strongest revenue streams.

2. Strengthen high-value customer retention

Introduce targeted loyalty programs, personalized offers, and retention campaigns for high-value and repeat customers rather than relying only on broad promotional campaigns.

3. Optimize discount strategy

Discounts should be evaluated based on **incremental revenue and profitability**, not simply order volume. Since discounted and non-discounted orders have nearly identical AOV, the business should identify discount levels that generate additional demand without unnecessarily reducing margins.

4. Improve regional delivery performance

Investigate operational causes behind slower delivery times in cities such as **Chandigarh, Hyderabad, and Ahmedabad**. Logistics capacity, fulfillment locations, courier performance, and delivery routes should be reviewed for these locations

5. Use regional customer segmentation

Develop region-specific customer strategies based on high-value customers and purchasing behavior. This can help improve customer engagement while allocating marketing budgets more efficiently.

6. Improve customer-data quality

Reduce **Unknown Customer ID** records through stronger customer identification and data-entry validation. Accurate customer IDs will improve revenue attribution, segmentation, retention analysis, and future customer lifetime-value analysis.

7. Monitor category and product concentration

Because revenue is strongly concentrated in Electronics and a small number of products, management should continue investing in these areas while identifying opportunities to improve weaker categories and reduce dependence on a limited number of products.

Conclusion

This analysis provided a comprehensive view of **e-commerce sales, customer behavior, product performance, pricing, discounts, and delivery operations** using SQL. The results show strong revenue contribution from Electronics and high-performing products, a substantial repeat-customer base, and significant variation in customer value and regional performance.

The analysis also identified opportunities to improve **customer retention, discount effectiveness, product strategy, delivery operations, and customer-data quality**. Overall, the project demonstrates how SQL-driven analysis can transform transactional data into **actionable business insights that support revenue growth, customer engagement, and operational improvement**.